Fully Logic and Evolutionary Computing

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Exercise 17

This exercise verifies the representation of a fuzzy set using its α-cuts.

$$A = \{(0.2, x_1), (0.4, x_2), (0.6, x_3), (0.8, x_4), (1, x_5)\}$$

Level set:

$$L(A) = \{0.2, 0.4, 0.6, 0.8, 1\}$$

Alpha-cuts:

α	A_α
0.2	$\{x_1, x_2, x_3, x_4, x_5\}$
0.4	$\{x_2, x_3, x_4, x_5\}$
0.6	$\{x_3, x_4, x_5\}$
0.8	$\{x_4, x_5\}$
1	$\{x_5\}$

Verification:

x	$\mu_A(x) = \sup\{\alpha : x \in A_\alpha\}$
x_1	0.2
x_2	0.4
x_3	0.6
x_4	0.8
x_5	1

Thus, $\mu_A(x) = \sup_\alpha \alpha A_\alpha(x)$ is satisfied.

Exercise 20

This exercise computes the composition of fuzzy relations using different operators.

$$\mu(x, z) = \max_{y \in Y} [\mu_c(x, y) \star \mu_{mb}(y, z)]$$

(a) Using minimum operator

(x, z)	Calculation	Result
(2, 4)	max{0, 0.4, 0.1}	0.4
(12, 4)	max{0, 0.4, 0.9}	0.9
(2, 8)	max{0, 0, 0.1}	0.1
(12, 8)	max{0, 0, 0.7}	0.7

(b) Using product operator

(x, z)	Calculation	Result
(2, 4)	max{0, 0.24, 0.1}	0.24
(12, 4)	max{0, 0.24, 0.9}	0.9
(2, 8)	max{0, 0, 0.07}	0.07
(12, 8)	max{0, 0, 0.63}	0.63

Exercise 23

This exercise evaluates fuzzy rule outputs using the minimum implication.

$$A = (50, 75, 100), \quad B = (2, 5, 8)$$

$$\mu_{B*}(y) = \min\{\mu_A(x'), \mu_B(y)\}$$

Membership values of A:

x'	$\mu_A(x')$
50	0
60	0.4
75	1
80	0.8
100	0

Resulting outputs:

x'	Output
50	0
60	Clipped at 0.4
75	Full membership $\mu_B(y)$
80	Clipped at 0.8
100	0
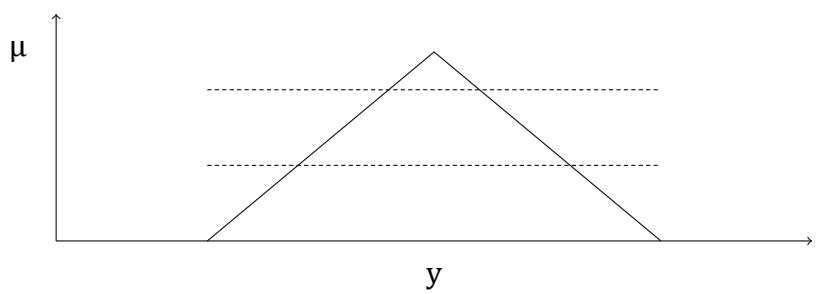
Figure: Output Membership Functions